A Negative Answer to the Separable Weak Banach Model Problem for $C_p[0,1]$

Automated Math Discovery Smoke Test

July 3, 2026

Source and Open Problem

This packet concerns Problem 3.10 from:

> Jerzy Kakol, Arkady Leiderman, and Artur Michalak, *A note on Banach spaces E admitting a continuous map from $C_p(X)$ onto E_w*, arXiv:2109.06338.

The problem is on page 12 of the source paper.

Remark 3.8. Necessary conditions in Theorem 1.7 are not sufficient, because the Banach space $E = \ell_1$ in the weak topology is never homeomorphic to any space $C_p(X)$. The reason is the following: every separable Banach space E with the Schur property is an $\aleph_0$-space in the weak topology. But $C_p(X)$ is an $\aleph_0$-space if and only if X is countable, i.e. if and only if $C_p(X)$ is metrizable. (For the details see [7]).

Remark 3.9. The arguments used in the proof of Theorem 1.7 are not applicable for the question whether X in that result must be a compact space and not just a σ-compact space. This is because compactness of X is not invariant under the homeomorphisms of the spaces $C_p(X)$. For instance, the spaces $C_p[0,1]$ and $C_p(\mathbb{R})$ are homeomorphic [9].

We finish the paper by the following challenging question.

Problem 3.10. *Does there exist a separable Banach space E such that E_w is homeomorphic to $C_p[0,1]$?*

References

[1] A. V. Arkhangel'ski, *Topological Function Spaces*, Kluwer, Dordrecht, 1992.
[2] C. S. Barroso, O. F. K. Kalenda, P. K. Lin, *On the approximate fixed point property in abstract spaces*, Math. Z. **271** (2012), 1271–1285.
[3] H. H. Corson, *The weak topology of a Banach space*, Trans. Amer. Math. Soc. **101** (1961),

In the notation of the source paper, E_w is the Banach space E endowed with its weak topology. The source asks:

> Does there exist a separable Banach space E such that E_w is homeomorphic to $C_p[0,1]$?

Answer

No. We prove a stronger statement: if K is any uncountable compact metrizable C-space, then $C_p(K)$ is not homeomorphic to E_w for any Banach space E. Since $[0,1]$ is a compact metrizable C-space, this settles Problem 3.10 negatively.

Proof Intuition

Krupski's proof that $C_p(K)$ is not homeomorphic to $C_w(L)$ for compact metrizable C-spaces K uses a finite-support analysis of a hypothetical homeomorphism. The coordinate functionals in that argument are the point masses δ_y on $C(L)$, indexed by a compact space L.

For a general Banach space E we replace the point masses by a compact Hilbert cube P of nonzero functionals in E^*. Each $p \in P$ gives the weak coordinate $e \mapsto p(e)$ on E_w. The finite-support machinery assigns to each p finitely many points of K controlling this coordinate through the homeomorphism. On a large strongly infinite-dimensional compact piece, these controlling points vary continuously. Since K is a C-space, dimension theory forces one fiber of the support map to be non-C.

The inverse homeomorphism supplies finitely many weak coordinates for each point of K. A Hahn–Banach separation argument then shows that every functional in the chosen fiber must lie in one fixed finite-dimensional linear span. A compact subset of a finite-dimensional space is a C-space, contradicting the non-C fiber.

Imported Topological Tool

We use the following standard finite-support decomposition theorem from the C_p literature. Krupski states the corresponding form for weak spaces $C_w(L)$ in [2, Theorem 1 and Lemma 1], following Marciszewski and van Mill. The proof uses only the compact metrizable parameter space and the continuity of the scalar coordinates; therefore the same statement applies to a compact parameter space $P \subset E^*$ with coordinates $e \mapsto p(e)$.

Lemma 1 (Finite-support decomposition)**.** *Let K and P be compact metrizable spaces, let Z be a space, and let $\Phi : C_p(K) \to Z$ be a homeomorphism. Suppose that for each $p \in P$ there is a continuous real-valued coordinate λ_p on Z, and that $p \mapsto \lambda_p(z)$ is continuous on P for each fixed $z \in Z$.*

For $k, m \in \mathbb{N}$, set

$$Z_{k,m} = \{(A, p) \in [K]^{\leq k} \times P : \lambda_p(\Phi(O_K(A; 1/m))) \subset [-1, 1]\},$$

where

$$O_K(A; 1/m) = \{f \in C_p(K) : |f(x)| < 1/m \text{ for all } x \in A\}.$$

Let $C(k,m)$ be the projection of $Z_{k,m}$ to P, and put $E(1,m) = C(1,m)$ and $E(k,m) = C(k,m) \setminus C(k-1,m)$ for $k > 1$.

If no coordinate $\lambda_p \circ \Phi$ is bounded on all of $C_p(K)$, then each $E(k,m)$ is covered by countably many G_δ subsets G of P such that on each G there are continuous maps

$$f_1, \dots, f_s : G \to K$$

with the following property. For every $p \in G$, the union of all minimal k-point sets $A \subset K$ satisfying $(A,p) \in Z_{k,m}$ is exactly

$$\alpha(p) = \{f_1(p), \dots, f_s(p)\}.$$

Main Theorem

Theorem 1. *Let K be an uncountable compact metrizable C-space. For every Banach space E, the spaces $C_p(K)$ and E_w are not homeomorphic.*

Proof. If E is finite-dimensional, then E_w is metrizable. Since K is uncountable, $C_p(K)$ is not metrizable. Thus only the infinite-dimensional case needs proof.

Assume toward a contradiction that there is a homeomorphism

$$\Phi : C_p(K) \longrightarrow E_w.$$

Composing with a translation in E_w, we may suppose $\Phi(0) = 0$.

A compact Hilbert cube of nonzero weak coordinates. Since E is infinite-dimensional, so is E^*. By the basic sequence theorem there is a normalized basic sequence $(u_n^*)_{n=0}^{\infty}$ in E^*. Put

$$P = \left\{ \frac{1}{2} u_0^* + \sum_{n=1}^{\infty} 2^{-n-4} t_n u_n^* : (t_n) \in [0,1]^{\mathbb{N}} \right\}.$$

The series converges uniformly in norm, so P is norm compact and hence w^*-compact. The coefficient representation is unique because (u_n^*) is basic; therefore P is homeomorphic to the Hilbert cube. Moreover every $p \in P$ is nonzero, since the perturbation has norm at most $\sum_{n=1}^{\infty} 2^{-n-4} = 1/16 < 1/2 = \left\| \frac{1}{2} u_0^* \right\|$. In particular P is a strongly infinite-dimensional compact metrizable space.

For $p \in P$, let $\lambda_p : E_w \to \mathbb{R}$ be the weak coordinate

$$\lambda_p(e) = p(e).$$

For fixed $e \in E$, the map $p \mapsto p(e)$ is w^*-continuous on P. Also $\lambda_p \circ \Phi$ is not bounded on $C_p(K)$, since Φ is onto E and p is a nonzero linear functional on E.

Closed finite-support relations. For $k, m \in \mathbb{N}$ define

$$Z_{k,m} = \{(A, p) \in [K]^{\le k} \times P : |p(\Phi(f))| \le 1 \text{ for every } f \in O_K(A; 1/m)\}.$$

Each $Z_{k,m}$ is closed. Indeed, if $(A, p) \notin Z_{k,m}$, choose $f \in O_K(A; 1/m)$ with $|p(\Phi(f))| > 1$. For A' close enough to A in the Vietoris topology, $A' \subset f^{-1}((-1/m, 1/m))$; and for p' close enough to p in the w^* topology, $|p'(\Phi(f))| > 1$. This gives an open neighborhood of (A, p) disjoint from $Z_{k,m}$.

Let $C(k, m)$ and $E(k, m)$ be as in Lemma 1. Since each $p \circ \Phi$ is continuous at 0, the sets $C(k, m)$ cover P. Applying Lemma 1, we obtain a countable cover of P by G_δ sets G_j such that on each G_j the minimal supports are enumerated by finitely many continuous maps into K.

A strongly infinite-dimensional support fiber. By the standard strongly infinite-dimensional compactum lemma used in [2, Proposition 2], if a strongly infinite-dimensional compact metrizable space is covered by countably many G_δ sets, then one of those sets contains a strongly infinite-dimensional compactum. Hence for some decomposition piece G there is a strongly infinite-dimensional compactum $Q \subset G$.

Let $f_1, \ldots, f_s : G \to K$ be the continuous support maps on this piece and define

$$F : Q \to K^s, \qquad F(p) = (f_1(p), \ldots, f_s(p)).$$

Finite products of compact metrizable C-spaces are C-spaces [4], so K^s is a C-space. If every fiber of F were a C-space, then Q would be a C-space by the standard Haver–Fedorchuk dimension theorem for compact metrizable spaces; this is impossible because Q is strongly infinite-dimensional. Therefore there exists $a = (a_1, \dots, a_s) \in K^s$ such that

$$Q_a = F^{-1}(a)$$

is not a C-space.

The inverse homeomorphism forces that fiber into a finite-dimensional span. Fix the integer m attached to the piece G. For every $x \in K$, continuity of the map

$$e \mapsto (\Phi^{-1}(e))(x)$$

from E_w to $\mathbb{R}$ at 0 gives a finite set $H_x^m \subset E^*$ and a number $\varepsilon_x > 0$ such that

$$e \in E, \quad |h(e)| < \varepsilon_x \text{ for every } h \in H_x^m \quad \Longrightarrow \quad |(\Phi^{-1}(e))(x)| < 1/m. \tag{1}$$

Set

$$N = \operatorname{span}\Big(\bigcup_{i=1}^{s} H_{a_i}^m\Big) \subset E^*.$$

This is finite-dimensional.

We claim that $Q_a \subset N$. Let $p \in Q_a$. By construction of the support decomposition, some minimal set $A \subset \alpha(p) = \{a_1, \dots, a_s\}$ satisfies

$$|p(\Phi(g))| \le 1 \quad \text{for every } g \in O_K(A; 1/m). \tag{2}$$

Suppose $p \notin N$. Since N is a finite-dimensional, hence w^*-closed, subspace of $(E^*, \sigma(E^*, E))$, the Hahn–Banach separation theorem gives $e \in E$ such that

$$p(e) = 2, \qquad h(e) = 0 \quad \text{for all } h \in N.$$

By (1), the function $g = \Phi^{-1}(e)$ satisfies $|g(a_i)| < 1/m$ for each $a_i \in A$, so $g \in O_K(A; 1/m)$. Then (2) gives $|p(e)| = |p(\Phi(g))| \le 1$, contradicting $p(e) = 2$. Thus $p \in N$, and the claim follows.

Now Q_a is a compact subset of the finite-dimensional space N endowed with the restricted weak-star topology. On finite-dimensional Hausdorff locally convex spaces all linear Hausdorff topologies coincide, so Q_a is a finite-dimensional compact metrizable space. In particular Q_a is a C-space. This contradicts the choice of Q_a as a non-C-space.

The contradiction proves the theorem. □

Corollary for Problem 3.10

Theorem 2. *There is no separable Banach space E such that E_w is homeomorphic to $C_p[0,1]$.*

Proof. Apply Theorem 1 to $K = [0,1]$, which is an uncountable compact metrizable C-space. □

Verification and Novelty Check

The proof is symbolic; no numerical computation is involved. The included asset-check script verifies that the packet files exist and that the source TeX contains the Problem 3.10 statement.

Bounded novelty check performed on July 3, 2026:

- searched the run indexes for `2109.06338`, $C_p[0,1]$, E_w, `separable Banach`, `weak topology`, and `homeomorphic`;
- searched the local parsed arXiv source corpus for exact and close phrases around `E_w is homeomorphic to C_p[0,1]` and `C_p([0,1]) weak topology separable Banach`;
- searched the web for the exact Problem 3.10 phrase and close variants.

The search found the source paper and related obstructions for spaces $C_w(L)$, notably [2], but no later paper explicitly answering Problem 3.10 or the stronger arbitrary-Banach formulation.

Limitations and Human Review Recommendation

The key review point is Lemma 1: it is the standard finite-support decomposition from the C_p homeomorphism literature, used with a compact parameter set P of nonzero weak coordinates in E^*. The rest of the proof is a direct separation and dimension-theoretic contradiction.

Recommended human verdict if Lemma 1 is accepted in this parameter form: 'full solution, likely publishable note'.

References

[1] J. Kakol, A. Leiderman, and A. Michalak, *A note on Banach spaces E admitting a continuous map from $C_p(X)$ onto E_w*, arXiv:2109.06338, 2021.

[2] M. Krupski, *On the weak and pointwise topologies in function spaces*, arXiv:1504.04554, 2015.

[3] J. van Mill, *The Infinite-Dimensional Topology of Function Spaces*, North-Holland Mathematical Library 64, North-Holland, Amsterdam, 2001.

[4] D. Rohm, *Products of infinite-dimensional spaces*, Proc. Amer. Math. Soc. **108** (1990), 1019–1023.

[5] F. Albiac and N. J. Kalton, *Topics in Banach Space Theory*, Graduate Texts in Mathematics 233, Springer, New York, 2006.